

ECE 361E: Machine Learning and Data Analytics for Edge AI

HW4 - In depth explanations

IMPORTANT! Do not use VSCode/PyCharm or any other IDE to connect to the edge devices in this class. They consume a lot of memory and impact everything that runs on the devices. Only use a plain terminal to connect to the devices.

In this homework, you should first *train your models exclusively* using the GPUs of Lonestar6, and then perform inference on the edge devices. **Use only the A100-Small nodes on Lonestar6.** You do *not* need to change the loss, the optimizer, the batch size, the number of training epochs or the learning rate when training. For time measurement, use the `time.time()` function.

IMPORTANT! All plots (in all homeworks) must have: a title, labels on both axes, a grid and a legend with corresponding labels for each line in the plot.

Use *matplotlib.pyplot* to draw figures and save your figures as PNG files. Save all data for the plots in a CSV file so there is no need to re-train in case of an error or system malfunctioning. For the entire HW4 use a sampling period of 200 milliseconds [ms] for power and temperature readings.

IMPORTANT! As a general rule for measuring power and temperature on edge devices, leave at least 2 minutes between running the benchmarks to allow the RaspberryPi 3B+ and Odroid MC1 thermals and loads to reach the idle steady-state. Of note, avoid using *print* statements in the measurement file since they may cause a small delay in the measurements/inference code that compounds over time.

For better readability, for all numbers that are over 999 in value, put comma signs, e.g., 1,800,903 instead of 1800903. Similarly, use 0.XXXX for decimal values (i.e., do not use more than 4 decimal digits) for any results written in tables, e.g., use 0.9384 instead of 0.938411291111. For time measurements, use the `time.time()` function.

The focus of this homework is primarily on the efficiency of the deployment (i.e., reducing latency and energy consumption), so you do not need to worry if you get a low accuracy **after** pruning, quantization or a combination of pruning and quantization. Use **Appendices A2.1 and A3.3** to connect to Raspberry Pi 3B+ and Odroid MC1 to deploy the models.

IMPORTANT! Everything you need is already provided to you on the devices. You are not allowed to install anything on the edge devices. By extension, you are not allowed to make any modification through software to the edge devices unless we mention it explicitly in the homework. When solving the homework, create your own directory and do not modify the *HW2_files*, *HW3_files* or *HW4_files* folders and their contents.

Problem 1: Structured Pruning

This problem deals with [Structured pruning](#) and uses the Torch-Pruning library to easily do this optimization for inference on Raspberry Pi 3B+.

For this problem do not use any other optimization for model compression except for pruning! When converting models to ONNX use the same functions you used in HW3.

Question 1

Do not modify the *main.py* file. If you run *main.py* as is, we expect you to reach a test accuracy greater than 70% for MobileNetV1. Feel free to use your Mobilenet-v1 checkpoint from HW3 as well.

The Torch-Pruning library prunes a certain fraction of the output channels for each convolutional layer of the given model. For instance, if the pruning fraction is set to 0.2 and we prune all convolutional layers, then the 20% of output channels with the lowest L1 norm will be pruned for each convolutional layer. To be more specific, if we have a convolutional layer with 50 output channels, then 10 output channels with the lowest L1 norm will be pruned.

To access certain modules (layers or blocks) in the MobileNetV1 model, use [*model.named_modules*](#) in PyTorch to iterate over the modules. **For this problem, fill out the TODO portions of the code, pass in the specified arguments to the script, and log your outputs to keep track of results.**

Question 2

For this question, you need to modify or add to the code from **Problem 1, Question1** to alter the pruning strategy. When choosing which pruning strategy is better, you need to consider both accuracy and number of parameters, e.g., for the same parameter count, which pruning version gets better accuracy.

Question 3

The latency and energy consumption values need to be averaged over the *entire* test dataset. Of note, you will run 7 models in total: one unpruned and 6 pruned with different pruning fractions. The power measurements for RaspberryPi 3B+ need to be done as in **HW2 and HW3**. Check **Appendix A3.5** for how to monitor memory consumption for RaspberryPi. Feel free to use the *deploy_onnx.py* and *get_power_temp.py* from HW3 for profiling.

For the *model size* column (Tables 1, 2, 3 of the HW4 document) use the size of the *.onnx* file on disk.

Problem 2: Static Post-Training Quantization

In this part of the homework, you will use static post-training quantization (PTQ) to optimize the deployment of MobileNet-v1 on RaspberryPi 3B+.

Question 1

For more details regarding quantization using the ONNX framework you should check [here](#); additionally, you can also check the **HW4 Demo video**.

Question 2

The latency and energy consumption values need to be averaged over the *entire* test dataset. You need to **deploy only the original (i.e., unpruned) MobileNet-v1 converted to ONNX with quantization**.

Question 3

Make sure to properly compare and contrast all important aspects you notice from your results in **Problem 1** and **Problem 2**, i.e., between using only pruning or using only quantization as optimization techniques for inference.

Problem 3: Inference Optimization

In this problem you need to use MobileNet-v1 with pruning and quantization. Finally, you can have fun with your own pruning+quantization optimization technique to obtain better results as a bonus problem.

Question 1

Here you must deploy all 6 pruned models deployed in **Problem 1, Question 3**, but converted to ONNX using PTQ.

Question 2

Note all the differences when combining structural pruning with static quantization over using pruning alone and quantization alone.

Question 3

You are required to use only the MobileNet-v1 model and optimize it *differently* than throughout **HW4**. Using what you learned during lectures and some inspiration from the research papers, properly motivate your optimization methodology and explain your choices.

Hint: You can use dynamic quantization, different pruning rates, smaller/larger number of fine-tuning epochs or change other hyperparameters to obtain a better optimization of latency, memory and energy values for inference, while having similar accuracy to previously optimized models.